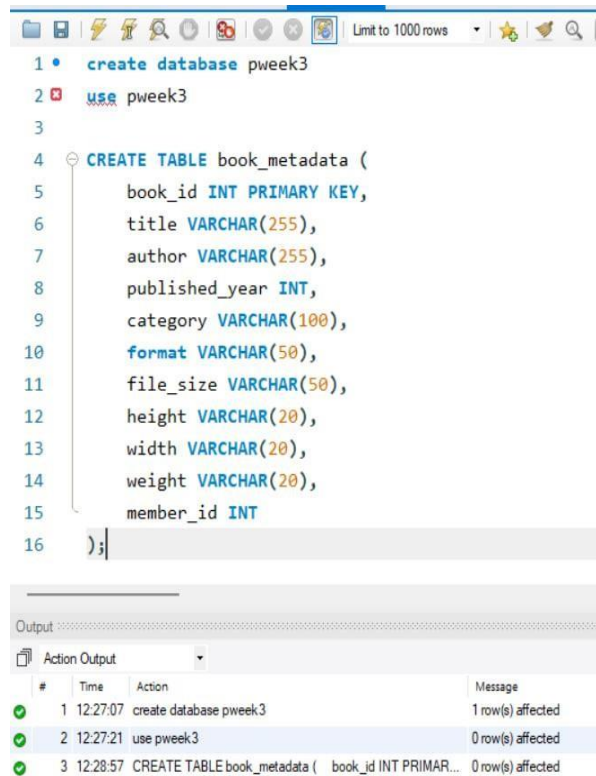


Practical week-3

Query 1: Create the table



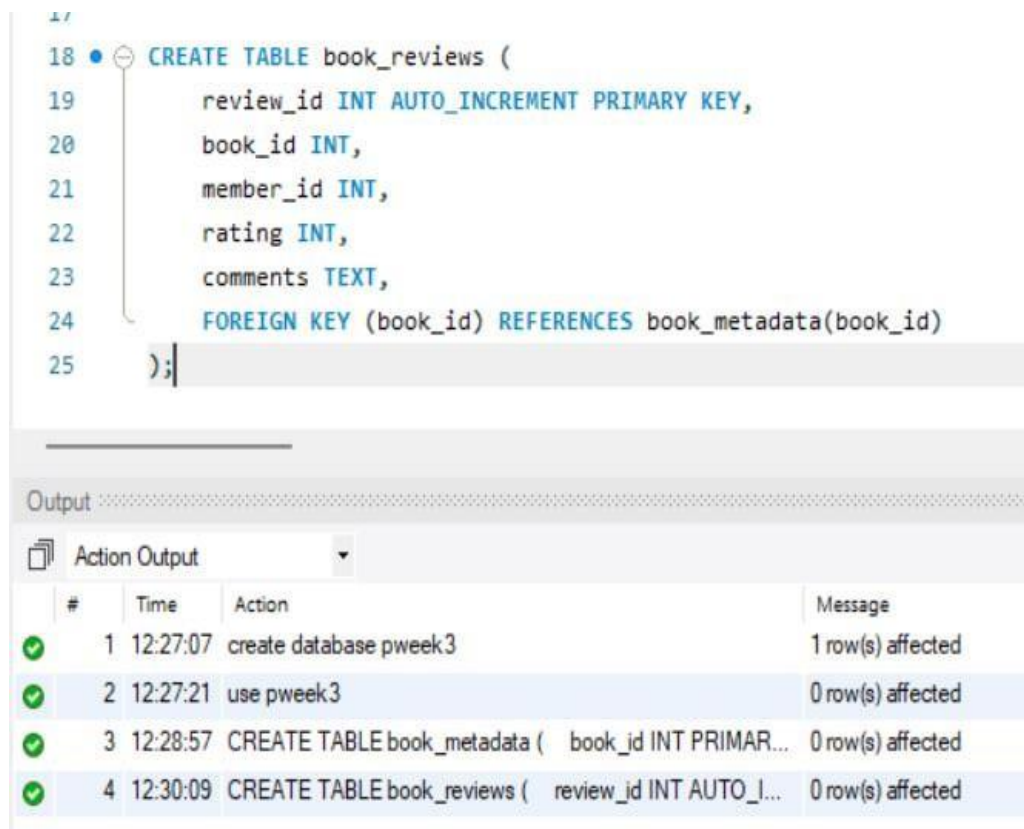
The screenshot shows a SQL query editor with the following code:

```
1 • create database pweek3
2 use pweek3
3
4 CREATE TABLE book_metadata (
5     book_id INT PRIMARY KEY,
6     title VARCHAR(255),
7     author VARCHAR(255),
8     published_year INT,
9     category VARCHAR(100),
10    format VARCHAR(50),
11    file_size VARCHAR(50),
12    height VARCHAR(20),
13    width VARCHAR(20),
14    weight VARCHAR(20),
15    member_id INT
16 );
```

Below the code, the 'Output' tab is selected, showing the 'Action Output' table:

#	Time	Action	Message
1	12:27:07	create database pweek3	1 row(s) affected
2	12:27:21	use pweek3	0 row(s) affected
3	12:28:57	CREATE TABLE book_metadata (book_id INT PRIMAR...	0 row(s) affected

Query 2: Create the book_reviews table



The screenshot shows a SQL query editor with the following code:

```
18 • CREATE TABLE book_reviews (
19     review_id INT AUTO_INCREMENT PRIMARY KEY,
20     book_id INT,
21     member_id INT,
22     rating INT,
23     comments TEXT,
24     FOREIGN KEY (book_id) REFERENCES book_metadata(book_id)
25 );
```

Below the code, the 'Output' tab is selected, showing the 'Action Output' table:

#	Time	Action	Message
1	12:27:07	create database pweek3	1 row(s) affected
2	12:27:21	use pweek3	0 row(s) affected
3	12:28:57	CREATE TABLE book_metadata (book_id INT PRIMAR...	0 row(s) affected
4	12:30:09	CREATE TABLE book_reviews (review_id INT AUTO_I...	0 row(s) affected

Query 3: Insert book metadata

```
40
27 • INSERT INTO book_metadata
28     (book_id, title, author, published_year, category, format, file_size,
29     VALUES
30     (101, 'The Martian', 'Andy Weir', 2021, 'Science Fiction',
31     'Digital PDF', '2.5 MB', NULL, NULL, NULL, 101),
32
33     (102, 'Dune', 'Frank Herbert', 2022, 'Science Fiction',
34     'Digital EPUB', '3.1 MB', NULL, NULL, NULL, 102),
35
36     (103, 'Clean Code', 'Robert C. Martin', 2019, 'Programming',
37     'Physical', NULL, '23 cm', '18 cm', '700 g', 103),
38
39     (104, 'Artificial Intelligence Basics', 'John Smith', 2023,
40     'Artificial Intelligence', 'Digital PDF', '4.2 MB',
41     NULL, NULL, NULL, 104),
42
43     (105, 'Data Structures', 'Mark Allen', 2024, 'Computer Science',
44     'Digital PDF', '5.0 MB', NULL, NULL, NULL, 105);
```

Output

Action Output

#	Time	Action	Message
✓ 1	12:27:07	create database pweek3	1 row(s) affected
✓ 2	12:27:21	use pweek3	0 row(s) affected
✓ 3	12:28:57	CREATE TABLE book_metadata (book_id INT PRIMAR...	0 row(s) affected
✓ 4	12:30:09	CREATE TABLE book_reviews (review_id INT AUTO_I...	0 row(s) affected
✓ 5	12:30:48	INSERT INTO book_metadata (book_id, title, publis...	5 row(s) affected Records: 5 Du

Query 4: Insert book reviews

```
• INSERT INTO book_reviews
  (book_id, member_id, rating, comments)
VALUES

(101, 101, 5, 'Excellent science fiction story'),
(101, 102, 4, 'Very interesting and informative'),
(101, 103, 5, 'Excellent space adventure'),

(102, 101, 5, 'Amazing world building'),
(102, 104, 4, 'Classic science fiction novel'),
(102, 105, 5, 'Excellent characters and story'),

(103, 101, 5, 'Excellent programming reference'),
(103, 106, 4, 'Very useful coding practices'),

(104, 102, 4, 'Good introduction to artificial intelligence'),
(104, 105, 5, 'Excellent machine learning explanation'),
(104, 106, 4, 'Useful for beginners'),

(105, 101, 5, 'Excellent data structures book'),
(105, 107, 3, 'Good but needs more examples');
```

Automatic correction disabled. Use the manual caret position toggle button.

Context Help Snippets

Action Output

Time	Action	Message
1 12:27:07	create database pweek3	1 row(s) affected
2 12:27:21	use pweek3	0 row(s) affected
3 12:28:57	CREATE TABLE book_metadata (book_id INT PRIMAR...	0 row(s) affected
4 12:30:09	CREATE TABLE book_reviews (review_id INT AUTO_I...	0 row(s) affected
5 12:30:48	INSERT INTO book_metadata (book_id, title, author, publis...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
6 12:31:32	INSERT INTO book_reviews (book_id, member_id, rating, c...	13 row(s) affected Records: 13 Duplicates: 0 Warnings: 0

Query 5: Select all records from book_metadata

67

68 • `SELECT * FROM book_metadata;`

Result Grid | Filter Rows: | Edit: | Export/Import:

	book_id	title	author	published_year	category
▶	101	The Martian	Andy Weir	2021	Science Fiction
	102	Dune	Frank Herbert	2022	Science Fiction
	103	Clean Code	Robert C. Martin	2019	Programming
	104	Artificial Intelligence Basics	John Smith	2023	Artificial Intelligence
	105	Data Structures	Mark Allen	2024	Computer Science
•	NULL	NULL	NULL	NULL	NULL

book_metadata 1 x Apply Revert Context Help Snippet

Output

Action Output

#	Time	Action	Message
✓ 2	12:27:21	use pweek3	0 row(s) affected
✓ 3	12:28:57	CREATE TABLE book_metadata (book_id INT PRIM...	0 row(s) affected
✓ 4	12:30:09	CREATE TABLE book_reviews (review_id INT AUTO...	0 row(s) affected
✓ 5	12:30:48	INSERT INTO book_metadata (book_id, title, author, pu...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 6	12:31:32	INSERT INTO book_reviews (book_id, member_id, rating...	13 row(s) affected Records: 13 Duplicates: 0 Warnings: 0
✓ 7	12:32:58	SELECT * FROM book_metadata LIMIT 0, 1000	5 row(s) returned

Query 6: Select all records from book_reviews

68 • `SELECT * FROM book_metadata;`

69

70 • `SELECT * FROM book_reviews;`

Result Grid | Filter Rows: | Edit: | Export/Import:

	review_id	book_id	member_id	rating	comments
▶	1	101	101	5	Excellent science fiction story
	2	101	102	4	Very interesting and informative
	3	101	103	5	Excellent space adventure
	4	102	101	5	Amazing world building
	5	102	104	4	Classic science fiction novel
	6	102	105	5	Excellent characters and story
	7	103	101	5	Excellent programming reference
	8	103	106	4	Very useful coding practices
	9	104	102	4	Good introduction to artificial intelligence
	10	104	105	5	Excellent machine learning explanation

book_reviews 2 x Apply

Output

Action Output

#	Time	Action	Message
✓ 3	12:28:57	CREATE TABLE book_metadata (book_id INT PRIM...	0 row(s) affected
✓ 4	12:30:09	CREATE TABLE book_reviews (review_id INT AUTO...	0 row(s) affected
✓ 5	12:30:48	INSERT INTO book_metadata (book_id, title, author, pu...	5 row(s) affected Reco
✓ 6	12:31:32	INSERT INTO book_reviews (book_id, member_id, rating...	13 row(s) affected Rec
✓ 7	12:32:58	SELECT * FROM book_metadata LIMIT 0, 1000	5 row(s) returned
✓ 8	12:33:26	SELECT * FROM book_reviews LIMIT 0, 1000	13 row(s) returned

Query 7: Books with rating greater than 4

```

72 • SELECT DISTINCT b.book_id, b.title, b.author
73 FROM book_metadata b
74 JOIN book_reviews r
75 ON b.book_id = r.book_id
76 WHERE r.rating > 4;

```

Result Grid

	book_id	title	author
▶	101	The Martian	Andy Weir
	102	Dune	Frank Herbert
	103	Clean Code	Robert C. Martin
	104	Artificial Intelligence Basics	John Smith
	105	Data Structures	Mark Allen

Result 3 x Read Only

Output

Action Output

#	Time	Action	Message
✓	4 12:30:09	CREATE TABLE book_reviews (review_id INT AUTO...	0 row(s) affected
✓	5 12:30:48	INSERT INTO book_metadata (book_id, title, author, pu...	5 row(s) affected Recc
✓	6 12:31:32	INSERT INTO book_reviews (book_id, member_id, rating...	13 row(s) affected Recc
✓	7 12:32:58	SELECT * FROM book_metadata LIMIT 0, 1000	5 row(s) returned
✓	8 12:33:26	SELECT * FROM book_reviews LIMIT 0, 1000	13 row(s) returned
✓	9 12:34:04	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned

Query 8: Books with rating IN (4, 5)

```

77
78 • SELECT DISTINCT b.book_id, b.title, b.author
79 FROM book_metadata b
80 JOIN book_reviews r
81 ON b.book_id = r.book_id
82 WHERE r.rating IN (4, 5);

```

Result Grid

	book_id	title	author
▶	101	The Martian	Andy Weir
	102	Dune	Frank Herbert
	103	Clean Code	Robert C. Martin
	104	Artificial Intelligence Basics	Robert C. Martin
	105	Data Structures	Mark Allen

Result 4 x Read Only

Output

Action Output

#	Time	Action	Message
✓	5 12:30:48	INSERT INTO book_metadata (book_id, title, author, pu...	5 row(s) affected Recon
✓	6 12:31:32	INSERT INTO book_reviews (book_id, member_id, rating...	13 row(s) affected Recc
✓	7 12:32:58	SELECT * FROM book_metadata LIMIT 0, 1000	5 row(s) returned
✓	8 12:33:26	SELECT * FROM book_reviews LIMIT 0, 1000	13 row(s) returned
✓	9 12:34:04	SELECT DISTINCT b book_id, b title, b.author FROM bo...	5 row(s) returned
✓	10 12:34:36	SELECT DISTINCT b book_id, b title, b.author FROM bo...	5 row(s) returned

Query 9: Books with Digital format

```
84 • SELECT *
85 FROM book_metadata
86 WHERE format LIKE '%Digital%';
```

	book_id	title	author	published_year	category
▶	101	The Martian	Andy Weir	2021	Science Fiction
	102	Dune	Frank Herbert	2022	Science Fiction
	104	Artificial Intelligence Basics	John Smith	2023	Artificial Intelligence
	105	Data Structures	Mark Allen	2024	Computer Science
*	NULL	NULL	NULL	NULL	NULL

book_metadata5 x

Apply

Output

Action Output

	#	Time	Action	Message
✓	6	12:31:32	INSERT INTO book_reviews (book_id, member_id, rating...	13 row(s) affected Rec
✓	7	12:32:58	SELECT * FROM book_metadata LIMIT 0, 1000	5 row(s) returned
✓	8	12:33:26	SELECT * FROM book_reviews LIMIT 0, 1000	13 row(s) returned
✓	9	12:34:04	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned
✓	10	12:34:36	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned
✓	11	12:35:10	SELECT * FROM book_metadata WHERE format LIKE '...	4 row(s) returned

Query 10: Average rating and total reviews

```

88 • SELECT
89     b.title,
90     ROUND(AVG(r.rating), 2) AS average_rating,
91     COUNT(r.review_id) AS total_reviews
92 FROM book_metadata b
93 JOIN book_reviews r
94 ON b.book_id = r.book_id
95 WHERE b.published_year > 2020
96 GROUP BY b.book_id, b.title
97 ORDER BY average_rating DESC;

```

Result Grid			
	title	average_rating	total_reviews
▶	The Martian	4.67	3
	Dune	4.67	3
	Artificial Intelligence Basics	4.33	3
	Data Structures	4.00	2

Result 6 × ⓘ

Output

Action Output

#	Time	Action	Message
✓ 7	12:32:58	SELECT * FROM book_metadata LIMIT 0, 1000	5 row(s) returned
✓ 8	12:33:26	SELECT * FROM book_reviews LIMIT 0, 1000	13 row(s) returned
✓ 9	12:34:04	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned
✓ 10	12:34:36	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned
✓ 11	12:35:10	SELECT * FROM book_metadata WHERE format LIKE '...	4 row(s) returned
✓ 12	12:36:04	SELECT b.title, ROUND(AVG(r.rating), 2) AS averag...	4 row(s) returned

Query 11: Create FULLTEXT index on comments

```

99 • CREATE FULLTEXT INDEX idx_comments
100 ON book_reviews(comments);

```

Output

Action Output

#	Time	Action
✓ 8	12:33:26	SELECT * FROM book_reviews LIMIT 0, 1000
✓ 9	12:34:04	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...
✓ 10	12:34:36	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...
✓ 11	12:35:10	SELECT * FROM book_metadata WHERE format LIKE '...
✓ 12	12:36:04	SELECT b.title, ROUND(AVG(r.rating), 2) AS averag...
⚠ 13	12:36:50	CREATE FULLTEXT INDEX idx_comments ON book_re...

Query 12: FULLTEXT search for 'excellent'

```

102 • SELECT *
103 FROM book_reviews
104 WHERE MATCH(comments)
105 AGAINST('excellent');

```

review_id	book_id	member_id	rating	comments
1	101	101	5	Excellent science fiction story
3	101	103	5	Excellent space adventure
6	102	105	5	Excellent characters and story
7	103	101	5	Excellent programming reference
10	104	105	5	Excellent machine learning explanation
12	105	101	5	Excellent data structures book
NULL	NULL	NULL	NULL	NULL

book_reviews 7 x Apply

Output

Action Output

#	Time	Action	Message
9	12:34:04	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned
10	12:34:36	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned
11	12:35:10	SELECT * FROM book_metadata WHERE format LIKE '...	4 row(s) returned
12	12:36:04	SELECT b.title, ROUND(AVG(r.rating), 2) AS averag...	4 row(s) returned
13	12:36:50	CREATE FULLTEXT INDEX idx_comments ON book_re...	0 row(s) affected
14	12:37:18	SELECT * FROM book_reviews WHERE MATCH(comm...	6 row(s) returned

Query 13: FULLTEXT search for 'science fiction'

```

107 • SELECT *
108 FROM book_reviews
109 WHERE MATCH(comments)
110 AGAINST('science fiction');

```

review_id	book_id	member_id	rating	comments
1	101	101	5	Excellent science fiction story
5	102	104	4	Classic science fiction novel
NULL	NULL	NULL	NULL	NULL

book_reviews 8 x Apply

Output

Action Output

#	Time	Action	Message
10	12:34:36	SELECT DISTINCT b.book_id, b.title, b.author FROM bo...	5 row(s) returned
11	12:35:10	SELECT * FROM book_metadata WHERE format LIKE '...	4 row(s) returned
12	12:36:04	SELECT b.title, ROUND(AVG(r.rating), 2) AS averag...	4 row(s) returned
13	12:36:50	CREATE FULLTEXT INDEX idx_comments ON book_re...	0 row(s) affected
14	12:37:18	SELECT * FROM book_reviews WHERE MATCH(comm...	6 row(s) returned
15	12:38:03	SELECT * FROM book_reviews WHERE MATCH(comm...	2 row(s) returned

Query 14: EXPLAIN FULLTEXT search

```
112 • EXPLAIN
113 SELECT *
114 FROM book_reviews
115 WHERE MATCH(comments)
116 AGAINST('excellent');
```

Result Grid							
		Filter Rows:			Export:	Wrap Cell Content: I A	
	id	select_type	table	partitions	type	possible_keys	key
▶	1	SIMPLE	book_reviews	NULL	fulltext	idx_comments	idx_comment

Result 9 ×					Read O
Output					
Action Output					
	#	Time	Action	Message	
✓	11	12:35:10	SELECT * FROM book_metadata WHERE format LIKE '...	4 row(s) returned	
✓	12	12:36:04	SELECT b.title, ROUND(AVG(r.rating), 2) AS averag...	4 row(s) returned	
⚠	13	12:36:50	CREATE FULLTEXT INDEX idx_comments ON book_re...	0 row(s) affected,	
✓	14	12:37:18	SELECT * FROM book_reviews WHERE MATCH(comm...	6 row(s) returned	
✓	15	12:38:03	SELECT * FROM book_reviews WHERE MATCH(comm...	2 row(s) returned	
✓	16	12:38:27	EXPLAIN SELECT * FROM book_reviews WHERE MAT...	1 row(s) returned	